

## Article

# Comparison of Machine Learning Models to Predict Nighttime Crash Severity: A Case Study in Tyler, Texas, USA

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**Abstract:** Driving at night is riskier in terms of crash involvement than it is during the day. Fortunately, it is clearly established that illumination on roadways can reduce the number and severity of nighttime crashes. However, state and municipal departments of transportation (DOTs) lack the available illumination data. Therefore, the objective of this research is threefold, as follows: (i) to develop machine learning models that use readily available roadway characteristic data to predict the severity of nighttime crashes; (ii) determine the effect that illumination has on crash severity; and (iii) develop a tool to assist DOT decision makers in collecting illumination data. To accomplish this objective, we have extracted data from the Texas Department of Transportation (TxDOT) Crash Record Information System (CRIS) database, which was then further split into a training and a test dataset. Then, seven machine learning techniques, namely binary logistic regression, k-nearest neighbors, naïve Bayes, random forest, artificial neural network, Extreme Gradient Boosting (XGBoost), and a Long Short-Term Memory (LSTM) model, were all applied to the unseen test data. The random forest model produced the most promising results by predicting severe crashes with 97.6% accuracy. In addition, we conducted a pilot study to test the collection of illumination data using a light meter. In the future, we aim to complete the development of a smartphone application, which can be used in conjunction with the random forest model presented in this paper, to collect crowdsourced illumination data and predict nighttime crash hotspots. This may assist DOT decision makers to prioritize funding for illumination at the hot spots.

**Keywords:** applied machine learning; crash severity; illumination; safety; nighttime crashes



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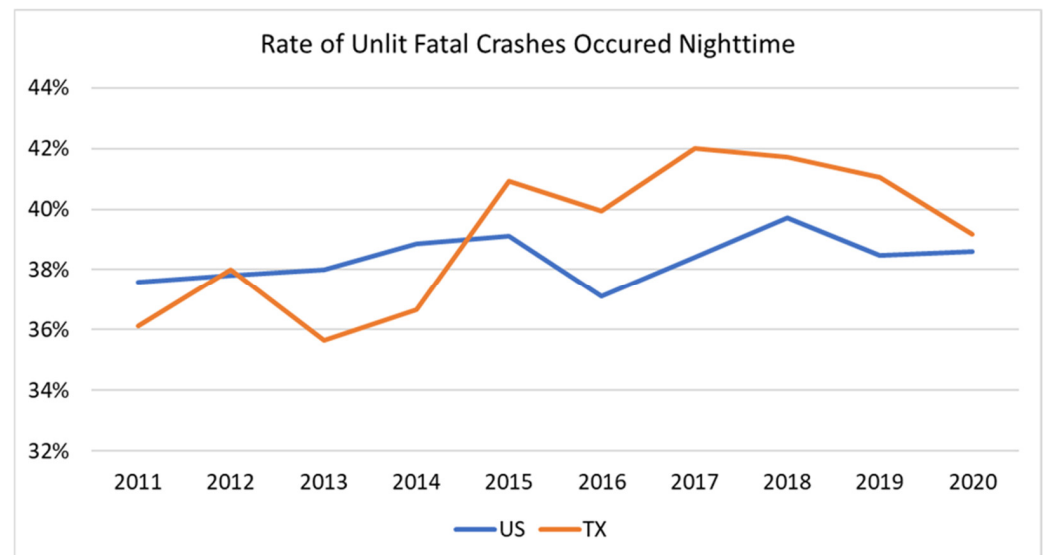
## 1. Introduction

Nighttime crashes have been shown to be riskier than daytime crashes in terms of crash involvement per distance traveled. Fatality rates are also three times higher at night than they are during the day, especially for crashes involving vulnerable road users (VRUs) such as pedestrians. In fact, pedestrians are seven times more likely to be involved in a fatal collision at night than they are during the day [1]. Understanding the visibility problems associated with nighttime driving is crucial for improving roadway safety.

Fortunately, the presence of illumination on roadways has been shown to reduce crash risk at night. Studies have found that installing streetlights on unlit roads can reduce

nighttime crashes by 55% [2]. Based on the National Highway Traffic Safety Administration (NHTSA) Fatality Analysis Reporting System (FARS) dataset [3], 4234 nighttime fatal crashes were reported in unlit urban areas in the U.S. in 2020. This is a significant increase from the number of crashes in 2013, which was only around 2700.

In Texas, the problem is particularly acute. In 2020, Texas experienced a significant increase of 518 fatal crashes reported in unlit locations on urbanized Texas roadways, which used to be only 325 in 2013 [3]. In fact, 40% of all nighttime fatal crashes in urbanized areas in Texas occurred at unlit locations. As demonstrated in Figure 1, the rate of unlit crashes is higher in Texas compared to the national average.



**Figure 1.** Rate of unlit fatal crashes occurring at night.

Given the clear link between insufficient illumination and nighttime roadway crashes, it is essential that agencies have enough data on roadway illumination. This way, unlit locations can be linked with other parameters, such as average annual daily traffic and number of lanes, so that available funding may be prioritized in order to reduce the number of nighttime crashes. However, based on current knowledge, there is no device to help agencies to collect, store, disseminate, and analyze the illumination infrastructure in their region. Therefore, it is imperative to develop tools and techniques that allow for efficient and effective collection of illumination data. This is where machine learning models come into play. By using readily available roadway characteristics, such as traffic volume and light condition, predictive models can be developed to determine the hotspots, the effects of illumination, and the crash severity. Therefore, to improve the understanding of the relationship between roadway illumination and nighttime crashes, the objective of this research was to answer the following questions:

- Research Question 1 (RQ1): Can readily available roadway and traffic volume characteristics be used as input parameters to predict the crash severity of nighttime crashes?
- Research Question 2 (RQ2): How significant is roadway illumination with respect to crash severity?
- Research Question 3 (RQ3): Can a tool (i.e., smartphone application) be developed to accurately measure roadway illumination?

The answer to RQ1 will inform transportation agencies as to whether they can predict severe crash hot spots using a prediction model. The answer to RQ2 will establish the significance of roadway infrastructure lighting in relation to safety and crash severity. The answer to RQ3 will determine the feasibility of developing a tool (i.e., smartphone

application) that can be used by drivers and/or agency maintenance staff to collect roadway illumination and geolocation data.

This article is organized as follows: After this introduction, past research on illumination and crash severity is reviewed. This is followed by a description of the Texas Department of Transportation (TxDOT) Crash Record Information System (CRIS) data that were extracted and used as part of this study. The subsequent sections describe the seven machine learning models that were applied, as well as a comparative evaluation. This paper concludes by highlighting the findings and contributions of this research, specifically as they relate to answering the three research questions.

## 2. Review of the Problem and Related Works

The United States Federal Highway Administration (FHWA) acknowledges the significance of nighttime safety issues on roadways. In their latest Every Day Counts initiative “Nighttime Visibility for Safety” is selected as an ongoing topic [4]. FHWA recognizes that nighttime fatality rates are three times higher than daytime rates and improving nighttime visibility at urban intersections with a focus on crosswalk visibility can significantly enhance safety. FHWA recommends using available tools and new approaches to improve nighttime visibility for all road users, including advancements in design, maintenance, and technology.

Numerous studies have examined the types of road lighting, including the technology and color preferences. One such study, conducted by Al Haji [5] in Sweden, focused on light-emitting diodes (LED) and adaptive lighting systems (ALS) used in roadway illumination. The study aimed to assess the impact of these technologies on traffic safety and risk for different road users, such as drivers and pedestrians. The findings suggest that it is essential to evaluate the safety implications of these technologies for individuals using the roadways. One possible methodology for doing so was introduced to carry out a pre- and post-implementation analysis, allowing for a precise comparison between these technologies and traditional road lighting systems. For this kind of analysis, it is also crucial to control other factors such as geometric design and traffic volume, which may impact safety levels. In another study, Park et al. [6] examined the impact of various lighting color temperatures on visibility distance in foggy conditions at night. The study involved a real-scale test with 91 participants. The results showed that road lighting with a low color temperature, specifically yellow, resulted in greater visibility distance compared to high color temperature lighting in conditions of night fog.

The research conducted by Bullough et al. [7] aimed to explore the theoretical connections between lighting, visibility, and safety using two distinct methodologies: statistical and analytical approaches. The study sought to examine the impact of lighting on crash frequency across various intersection types in Minnesota. Count regression models were employed to evaluate the findings, which revealed that the existence of illumination at roadway intersections is linked to a decrease of about 12% in the ratio of night-to-day crashes compared to unlit intersections.

Donnell et al. [8] developed a framework for estimating the safety impacts of fixed lighting across diverse intersection types and locations. The study utilized data from the Highway Safety Information System (HSIS), which is designed to investigate the correlation between intersection lighting and three distinct safety performance metrics, as follows: the anticipated frequency of daytime crashes, the projected frequency of nighttime crashes, and the projected percentage shift in the night-to-day crash ratio.

The research conducted by Bhagavathula et al. [9] aimed to investigate the correlation between the level of roadway illuminance and nighttime rural intersection safety. The study analyzed the impact of both lighting level and quality on night-to-day crash ratios

at rural intersections, using the roadway crash database of the Virginia Department of Transportation as the source of the crash data. Lighting data were obtained through the use of a mobile measurement system for roadway lighting. The study found that the average horizontal illuminance referring to lighting level has a significant impact on rural intersection safety at night. The results indicated that unlit intersections have higher night-to-day crash ratios compared to lighted intersections.

Yu et al. [10] developed safety performance functions (SPFs) to predict nighttime crashes in relation to roadway lighting. The number of lanes played a crucial role in the context of interchange ramps. It was observed that entry ramps had considerably higher crash rates than exit ramps in situations where lighting was absent. Additionally, Johnson et al. [11] worked on developing an Advanced Lighting Measurement System (ALMS) that has the capability to capture illumination measurements at typical driving velocities.

Recent advancements in machine learning have been applied to different fields of transportation safety, including crash prediction. In the UK, a 2023 study by Sufian and Varadarajan [12] enhanced the prediction of UK road traffic crash severity by integrating machine learning with econometric and time series forecasting methods, utilizing models like random forests and Extreme Gradient Boosting (XGBoost). Another 2023 study in Saudi Arabia compared multiple machine learning algorithms, including logistic regression, k-nearest neighbor, and decision trees, to identify the most effective model for predicting crash severity [13]. It was found that the decision tree and random forest models performed the best and that the most important factors were the type of crash followed by the time of the crash. Also in 2023, Huang et al. [14] developed machine learning models and provided explanations of how the model makes certain predictions based on input parameters such as road conditions, traffic volume, and weather. The explanation helps to interpret the machine learning models' prediction, which enables stakeholders to make informed decisions.

Previous studies in this context have discussed the impact of road lighting on crash frequency and the importance of improving nighttime visibility for roadway safety. While previous research has explored some of the theoretical connections between lighting, visibility, and safety, there is still a lack of available data and novel techniques to predict crash severity. This study aims to develop machine learning predictive models to predict crash severity and to assist transportation agency decision makers in collecting illumination data using a smartphone application. In short, this study aims to fill the research gap by recommending a data collection technique, as well as predictive models, to determine the nighttime crash hotspots with poor illumination.

### 3. Crash Severity Parameters

There are numerous factors that can contribute to a crash, which include the driver, the vehicle(s), and the roadway conditions (i.e., weather, geometric layout of the road, etc.). In order to develop a crash severity prediction model, readily available input parameters are needed. Therefore, for this research, the input parameters related to crash severity that were considered are as follows:

- Average Annual Daily Traffic Amount, AADT ( $\frac{\text{vehicles}}{\text{day}}$ );
- Percentage of Average Daily Traffic for Trucks (% of AADT);
- Light Condition (1 if lighted conditions, 0 otherwise);
- Number of Lanes;
- Percentage of Combo Truck Average Daily Traffic (% of AADT);
- Percentage of Single Unit Truck Average Daily Traffic (% of AADT);
- Right Shoulder Width (feet);
- Roadbed Width (feet);
- Speed Limit (mph);

- Surface Condition (1 if dry conditions, 0 otherwise);
- Surface Width (feet);
- Weather Condition (1 if clear weather, 0 otherwise).

The output parameter is the crash severity. If a crash yielded at least one major injury and/or a fatality, then the crash was said to be severe. Otherwise, the crash was considered non-severe. These input parameters are all readily available to a DOT official through their roadway database and a weather monitor. The idea was, by choosing readily available parameters that are easily obtainable, the parameters that contribute the most to crash severity can be discovered so that DOT officials may predict the following: (i) the roadways that will yield more severe crashes; and (ii) the contributing factors so that proper countermeasures may be put in place to reduce the severity of future crashes.

#### 4. Data Description

For this research, the data used were extracted from the Texas Department of Transportation (TxDOT) Crash Record Information System (CRIS) database [15]. The CRIS database contains crash data in Texas for the past ten years. Therefore, the crash data used for this research were from the period of 2013 to 2023. One of the constraints when querying the CRIS data is that no more than 50,000 data points may be extracted at once. Therefore, to limit the scope of this research, we focused on crashes primarily in Smith County (i.e., the county where Tyler, Texas, is located).

The CRIS database was filtered to only include crashes that occurred at night over the past ten years. This was carried out so that the effect of illumination could be determined. Crashes during the day, dusk, or dawn were not considered as part of this research. From there, the data were further processed as follows:

- First, the CRIS database categorizes crashes based on severity using categorical values, as follows [15]:
  - K: fatal, fatal injury, death within 30 days of a crash;
  - A: disabling, suspected serious injury, incapacitating injury;
  - B: evident, suspected minor injury, non-incapacitating injury;
  - C: possible injury (no visible evidence but the person claimed that he/she was injured);
  - O: no injury;
  - PDO: property damage only;
  - Unknown: no person involved was present at the scene during the investigation.
- For any unknown severity of crashes, these values were removed. Next, fatal and disabling serious injuries were then assigned values of 1. All other crashes were assigned values of 0. Any crash that had any missing values for any of the input parameters was excluded. This was carried out so that the models may be developed and comparatively evaluated without missing values. As part of future research, missing data may be presented to the selected model to compare the model's performance as part of a fault tolerance analysis. Past research in transportation engineering has studied the fault tolerance of developed machine learning models [16].
- Lastly, the parameters of light condition, surface condition, and weather condition were altered from categorical values to binary values as follows:
  - Light Condition: 1 if lighted conditions, 0 otherwise;
  - Surface Condition: 1 if dry conditions, 0 otherwise;
  - Weather Condition: 1 if clear weather, 0 otherwise.

- This was carried out so that mathematical and statistical techniques may be performed and so that the effect of lighting, surface, and weather conditions could be determined. Once all of the data had been processed, there were 10,500 crashes in total.

The descriptive statistics for the extracted crash dataset are presented in Table 1. Since the objective of this research is to apply machine learning techniques to be used as a predictive model, the overall data presented in Table 1 were separated into a training and test dataset using a stratified 80/20 random sampling split. Table 2 describes the processed data.

**Table 1.** Descriptive statistics of extracted data from CRIS.

| Parameter                                             | Unit                                    | Sample Size | Min | Max    | Mean   | Std. Deviation |
|-------------------------------------------------------|-----------------------------------------|-------------|-----|--------|--------|----------------|
| AADT                                                  | $\frac{\text{vehicles}}{\text{day}}$    | 10,500      | 168 | 42,065 | 21,087 | 12,643         |
| Percentage of Average Daily Traffic for Trucks        | %                                       | 10,500      | 2   | 33     | 11     | 9              |
| Light Condition                                       | One if lighted at crash,<br>0 otherwise | 10,500      | 0   | 1      | 0.45   | 0.50           |
| Number of Lanes                                       | lanes                                   | 10,500      | 2   | 6      | 4      | 2              |
| Percentage of Combo Truck Average Daily Traffic       | %                                       | 10,500      | 0   | 28     | 7      | 8              |
| Percentage of Single Unit Truck Average Daily Traffic | %                                       | 10,500      | 2   | 13     | 4      | 1              |
| Right Shoulder Width                                  | ft                                      | 10,500      | 0   | 20     | 6      | 7              |
| Roadbed Width                                         | ft                                      | 10,500      | 20  | 126    | 68     | 23             |
| Speed Limit                                           | mph                                     | 10,500      | 10  | 80     | 53     | 13             |
| Surface Condition                                     | One if dry surface,<br>0 otherwise      | 10,500      | 0   | 1      | 0.78   | 0.42           |
| Surface Width                                         | ft                                      | 10,500      | 8   | 126    | 58     | 24             |
| Weather Condition                                     | One if clear weather,<br>0 otherwise    | 10,500      | 0   | 1      | 0.84   | 0.36           |
| Crash Severity                                        | One if severe crash,<br>0 otherwise     | 10,500      | 0   | 1      | 0.36   | 0.48           |

**Table 2.** Overview of training and test datasets.

| Overall Dataset                                          |        |
|----------------------------------------------------------|--------|
| Number of total crashes                                  | 10,500 |
| Number of crashes with an output of 0 (non-severe crash) | 6734   |
| Number of crashes with an output of 1 (severe crash)     | 3766   |
| Training Dataset                                         |        |
| Number of total crashes                                  | 8400   |
| Number of crashes with an output of 0 (non-severe crash) | 5387   |
| Number of crashes with an output of 1 (severe crash)     | 3013   |
| Test Dataset                                             |        |
| Number of total crashes                                  | 2100   |
| Number of crashes with an output of 0 (non-severe crash) | 1347   |
| Number of crashes with an output of 1 (severe crash)     | 753    |

It can be observed from Table 2 that, out of the 10,500 total crashes that occurred at night, approximately 36% were considered to be severe crashes. This is true for the overall dataset, as well as the training and test dataset, as the data were randomly split using a

stratified random sampling split. This may also be seen in Table 1, where the mean value for crash severity is 0.36, which indicates that approximately 36% of crashes were severe (i.e., crashes that yielded at least one major injury and/or a fatality).

## 5. Prediction Models

### 5.1. Application of Machine Learning Methods

After partitioning the data, various popular machine learning methods can be applied to the training dataset and their performance assessed using the unseen test dataset. Given that the outcome is binary, representing  $\{0, 1\}$  for  $\{\text{non-severe crash}, \text{severe crash}\}$ , respectively, only binary classification methods are suitable. Therefore, the seven machine learning techniques chosen for prediction models are as follows: binary logistic regression (BLR), k-nearest neighbor (KNN), naïve Bayes (NB), random forest (RF), artificial neural network (ANN), Extreme Gradient Boosting (XGBoost), and a Long Short-Term Memory (LSTM) model.

### 5.2. Performance Metrics

Each crash in the training and test datasets was labeled  $CS = 0$  or  $CS = 1$ , indicating *Crash Severity*, with 0 representing a non-severe crash and 1 representing a severe crash. When the input parameters were fed into one of the seven machine learning models, each model's output was indicated as *MD* (for *Model's Decision*), where  $MD = 0$  meant “not a severe crash”, and  $MD = 1$  meant “severe crash”. When  $MD = 0$ , this would inform DOT officials that, based on the roadway parameters and weather conditions, this roadway segment may not yield severe crashes; however, when  $MD = 1$ , this would inform DOT officials that, based on the roadway parameters and weather conditions, this roadway segment may yield severe crashes and should thus be considered further for safety countermeasures. Table 3 shows the confusion matrix of the test.

**Table 3.** Confusion matrix.

|                                | $MD = 0$<br>[Not a Severe Crash]                                                             | $MD = 1$<br>[Severe Crash]                                                                           |
|--------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| $CS = 0$<br>[non-severe crash] | Correct decision<br>[True negative]                                                          | Wrong decision, may cause DOT officials to implement unnecessary countermeasures<br>[False positive] |
| $CS = 1$<br>[severe crash]     | Wrong decision, may cause DOT officials to overlook severe crash hotspot<br>[False negative] | Correct decision<br>[True positive]                                                                  |

The outcomes presented in Table 3 align with a binary classification problem and are evaluated using the following traditional performance metrics:

$$\text{Precision} = \frac{\text{No.of vectors with } [MD = 1|CS = 1]}{\text{No.of vectors with } [MD = 1]} \quad (1)$$

$$\text{Recall} = \frac{\text{No.of vectors with } [MD = 1|CS = 1]}{\text{No.of vectors with } [CS = 1]} \quad (2)$$

$$F1 \text{ score} = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}} \quad (3)$$

Precision measures the proportion of correct  $[MD = 1]$  decisions made by the models. Recall assesses the fraction of  $[CS = 1]$  vectors in the test dataset that are correctly identified

by the models. The F1 score is the harmonic mean of the precision and recall. In addition, we include the overall accuracy, or *OA*, as follows:

$$OA = \frac{\text{No. of vectors with } \{[MD = 1|CS = 1] + [MD = 0|CS = 0]\}}{\text{Total no. of vectors}} \quad (4)$$

*OA* measures the proportion of correct predictions out of all predictions.

These metrics were used to evaluate the performance of each model. Given that the consequence of a false negative (i.e.,  $[MD = 0|CS = 1]$ ) is more severe than a false positive (i.e.,  $[MD = 1|CS = 0]$ ), precision, recall, and F1 score are prioritized over *OA* in the comparative evaluation of the machine learning models.

### 5.3. Optimal Threshold Value for Each Model

Post-training, the models produced outputs within the continuous range of  $(0, 1)$ ; however, to create a confusion matrix, each *MD* vector needed to be binary  $\{0, 1\}$ . Therefore, a threshold value,  $\tau$ , was established such that  $MD = 0$  if the output was less than or equal to  $\tau$ , otherwise,  $MD = 1$ . To find the optimal  $\tau$  for each model, nine threshold values were tested, ranging from 0.1 to 0.9, in 0.1 increments. By plotting the performance curves (Equations (1)–(4)) of each model's accuracy against  $\tau$ , the optimal  $\tau$  value that maximized each model's accuracy could be identified. This threshold was identified for all machine learning models to achieve optimal performance.

### 5.4. Binary Logistic Regression (BLR)

The first machine learning technique utilized was binary logistic regression (BLR), a special form of linear regression where the dependent variable is binary. BLR is among the most straightforward and commonly used statistical methods for predicting binary outcomes. However, BLR tends to fall short in producing robust and accurate results, especially when dealing with asymmetrical data, like the one utilized in this study [17]. IBM SPSS Statistics 28 [18] was used to apply BLR to the training data in this study. The coefficients, illustrated in Table 4, shed light on the significance of each input parameter with respect to crash severity.

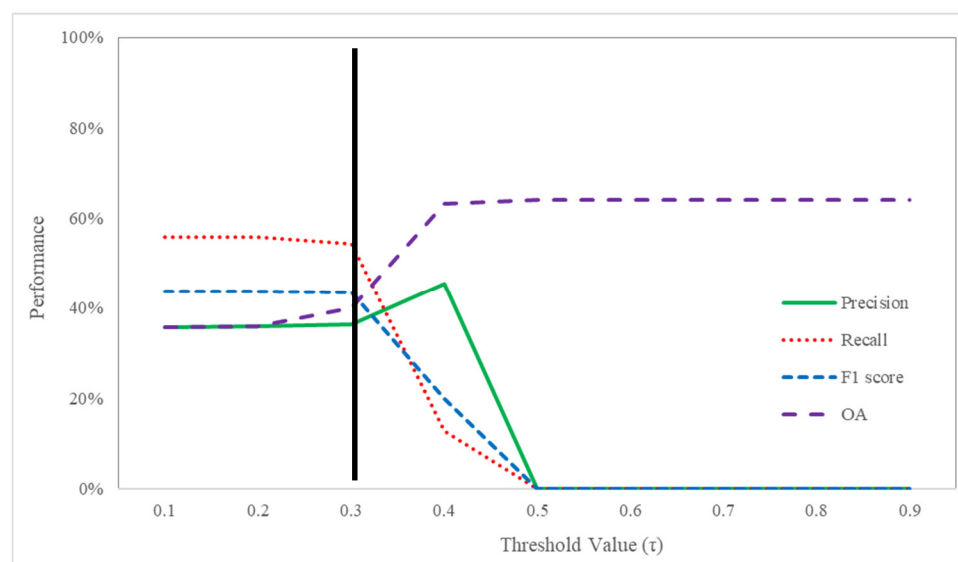
**Table 4.** Binary logit model of crash severity.

| Parameters                                            | Coefficient | Standard Error | <i>p</i> Value |
|-------------------------------------------------------|-------------|----------------|----------------|
| AADT                                                  | −0.000002   | 0.000004       | 0.591088       |
| Percentage of Average Daily Traffic for Trucks        | −1.233899   | 0.489628       | 0.011733       |
| Light Condition                                       | −0.089488   | 0.057961       | 0.122604       |
| Number of Lanes                                       | −0.034176   | 0.043867       | 0.435924       |
| Percentage of Combo Truck Average Daily Traffic       | 1.225178    | 0.489511       | 0.012319       |
| Percentage of Single Unit Truck Average Daily Traffic | 1.281559    | 0.491842       | 0.009170       |
| Right Shoulder Width                                  | −0.017728   | 0.015402       | 0.249731       |
| Roadbed Width                                         | 0.015575    | 0.009367       | 0.096372       |
| Speed Limit                                           | −0.001932   | 0.002557       | 0.449985       |
| Surface Condition                                     | 0.149660    | 0.089312       | 0.093797       |
| Surface Width                                         | −0.007415   | 0.009579       | 0.438888       |
| Weather Condition                                     | 0.097545    | 0.102558       | 0.341543       |
| Constant                                              | −1.083060   | 0.201816       | 0.000000       |

From the BLR results, the most significant parameters are the constant, the percentage of trucks in the average daily traffic, followed by roadbed width and surface condition. The light condition parameter has a  $P$  value of approximately 0.123. While this is not close enough to be considered statistically significant with an  $\alpha = 0.05$ , it could still be interpreted as providing some evidence against the null hypothesis. Furthermore, the coefficient for illumination is  $-0.089488$ , which makes sense intuitively in that, if illumination is added, the severity of crashes will decrease. This essentially answers the second research question, which asked if illumination is significant with respect to nighttime crash severity.

The coefficients for single unit and combo truck percentages are both positive, which makes sense intuitively, as more trucks would cause more severe crashes. In addition, the road width coefficient is positive, which indicates that as the width of the road increases, the crash severity will also increase. The surface condition coefficient is also positive, which indicates that if the surface condition is changed from anything other than a dry surface (i.e., wet, icy, etc.), then the severity of a crash will also increase.

After training the BLR model, it was applied to the unseen test dataset nine times, each with a different threshold value,  $\tau$ , ranging  $[0.1, 0.9]$  in 0.1 increments. The results of the performance curves are presented in Figure 2.



**Figure 2.** BLR performance curves from the test dataset.

At a low threshold of 0.1, any BLR model prediction greater than 0.1 is then classified as a severe crash (i.e.,  $MD = 1$ ). This can be seen by relatively high precision, recall, and F1 scores; however, the OA is relatively low because the BLR predictions are almost all severe crashes (i.e.,  $MD = 1$ ). As the threshold then increases, the BLR makes more  $MD = 0$  predictions. This can be seen by a decrease in precision, recall, and F1 score; however, conversely, the OA increases. Therefore, the threshold that makes the most correct decisions should be chosen as the optimal threshold. As presented in Figure 2, the optimal threshold value, which maximizes precision, recall, F1 score, and OA, was determined to be 0.3 (indicated by the black line). Therefore, a BLR with  $\tau = 0.3$  will be comparatively evaluated with the other machine learning techniques.

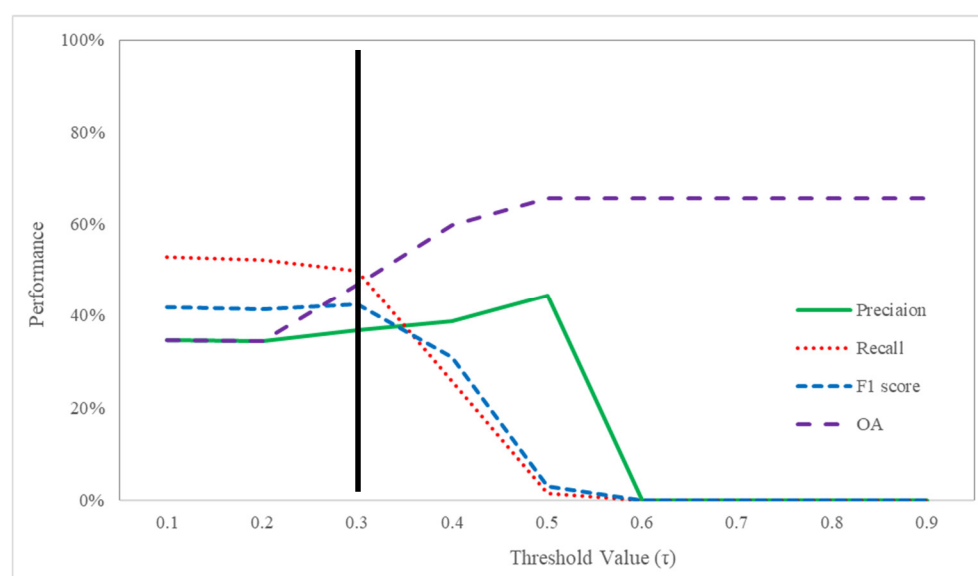
### 5.5. K-Nearest Neighbor (KNN)

K-nearest neighbor (KNN) is a widely used and straightforward classification technique after BLR. This non-parametric method classifies test points based on the Euclidean distance to training samples. For a given test point ( $x_0$ ), the classification is determined by the majority vote among its  $K$  neighbors ( $x_{(n)}, n = 1, \dots, K$ ). In cases of ties, a random

decision is made [19]. For the simplest scenario, where  $K = 1$  (which is termed as NN), the test sample is assigned the class of its closest neighbor. KNN is instance-based learning, meaning that it does not build a model but instead compares the new test data to training instances and defers computations until classification [20].

KNN does not assume any specific data distribution, and its performance is significantly influenced by the choice of  $K$  [21]. A very small value of  $K$  can lead to poor performance if the data are sparse or noisy, as the local region for classification is too limited. Conversely, a very large  $K$  may reduce accuracy due to the inclusion of outliers from neighboring classes.

IBM SPSS Statistics 28 [18] was used to apply KNN to the training data. Various values of  $K$  were tested, and the optimal number of neighbors was found to be 88. Using this optimal  $K$ , the KNN model was then applied to the unseen test data to determine the best threshold value,  $\tau$ . Similar to BLR, a threshold of 0.3 yielded the best performance results. The performance curves are depicted in Figure 3.



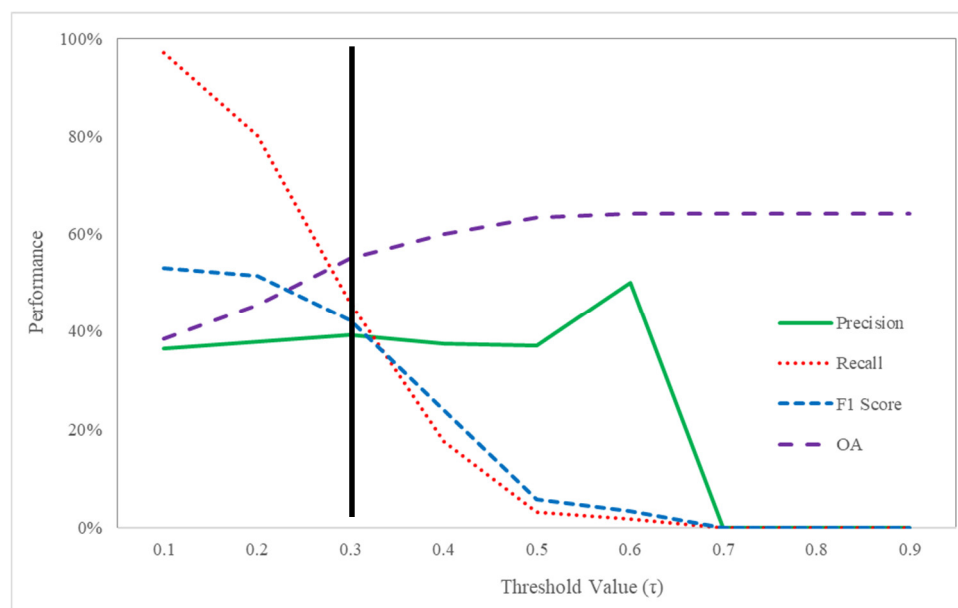
**Figure 3.** KNN performance curves from the test dataset.

#### 5.6. Naïve Bayes (NB)

Naïve Bayes (NB) is the third classifier tested in this research, which is a popular and efficient machine learning algorithm used for classification tasks. It is based on Bayes' Theorem and assumes that all features are independent of each other, hence the name "naive" [22]. This classifier calculates the probability of a given instance belonging to a particular class by combining prior knowledge (prior probability) and the probability of observing the features given, such as class (likelihood).

For the NB model, we used the Matlab Classification Learner application [23]. The only hyperparameter that may be tuned is the kernel type, which is the distribution of each input parameter. The available kernel distribution hyperparameters that were tuned were the Gaussian, box, Epanechnikov, and triangle, all of which were tuned using the random grid search algorithm and a fivefold cross validation. The optimal kernel distribution was the triangle distribution. Figure 4 depicts the performance curves for the tuned NB model on the unseen test dataset.

The tuned NB model had a wide range of predicted values within the range of [0.1, 0.7]. According to the results shown in Figure 4, the optimal threshold that maximizes performance on the unseen test data is  $\tau = 0.3$ .



**Figure 4.** NB performance curves from the test dataset.

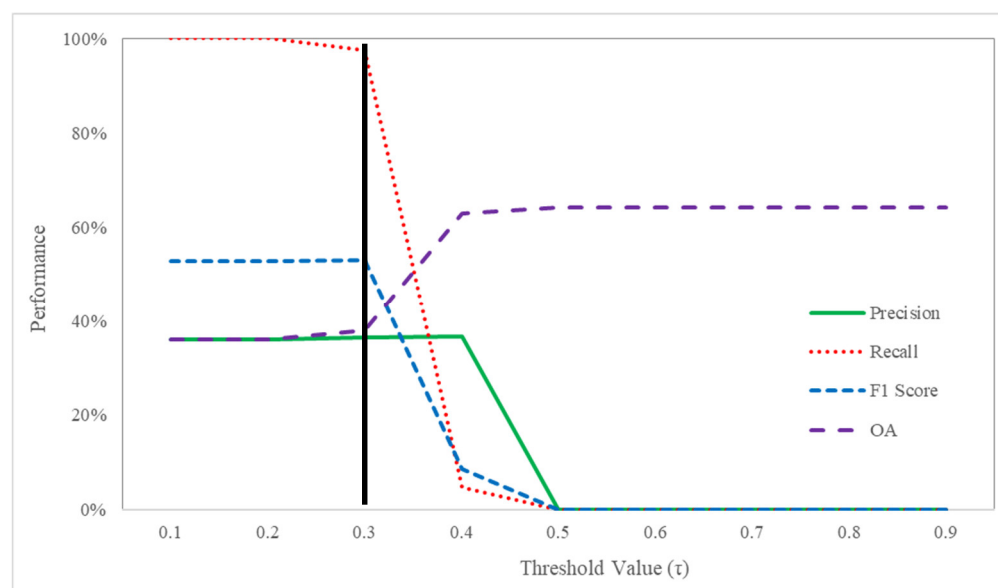
### 5.7. Random Forest (RF)

Random forest (RF) is the fourth technique utilized, and it represents an advanced form of bagging (or bootstrap aggregation) that combines the results from a multitude of uncorrelated decision trees [24]. This ensemble method relies on multiple decision trees, where each tree votes on the classification, and the class with the majority of votes is chosen as the RF model's prediction. This approach typically has lower bias if the trees are sufficiently deep, though it can exhibit high variance, as small changes in the training data may lead to significantly different trees.

For a training dataset  $M$  with multiple data points  $(x_1, y_1) \dots (x_n, y_n)$ , where  $x_i$  is the training vector and  $y_i$  is the output class for  $x_i$ , the RF algorithm first creates numerous decision trees, each constructed from a bootstrap sample of the training data. In this process, sampling is performed with replacement, so some training instances may be included multiple times while others may not be included at all. Each tree is built to its maximum depth without pruning, using randomly selected features at each split until the terminal nodes contain only a single class or reach a minimum size [25]. Finally, the votes are averaged over each tree in the forest to obtain the most likely class.

For this model, we used the Matlab Classification Learner application [23]. The two hyperparameters that were tuned were the maximum number of splits and the number of learners using the random grid search algorithm and a fivefold cross validation. The optimal number of splits was four and the optimal number of learners was 63. More information about the bagging ensemble method and random forests is described extensively in [24]. Figure 5 presents the various evaluation metrics for the tuned RF model on the unseen test data.

A majority of the RF model's predictions for the unseen test data fell within the range of  $[0.3, 0.4]$ , which is illustrated by drastic increases and decreases in the performance curves between these two threshold values. Since the recall is nearly perfect when using a threshold,  $\tau = 0.3$ , that threshold will be used as part of the comparative evaluation.



**Figure 5.** RF performance curves from the test dataset.

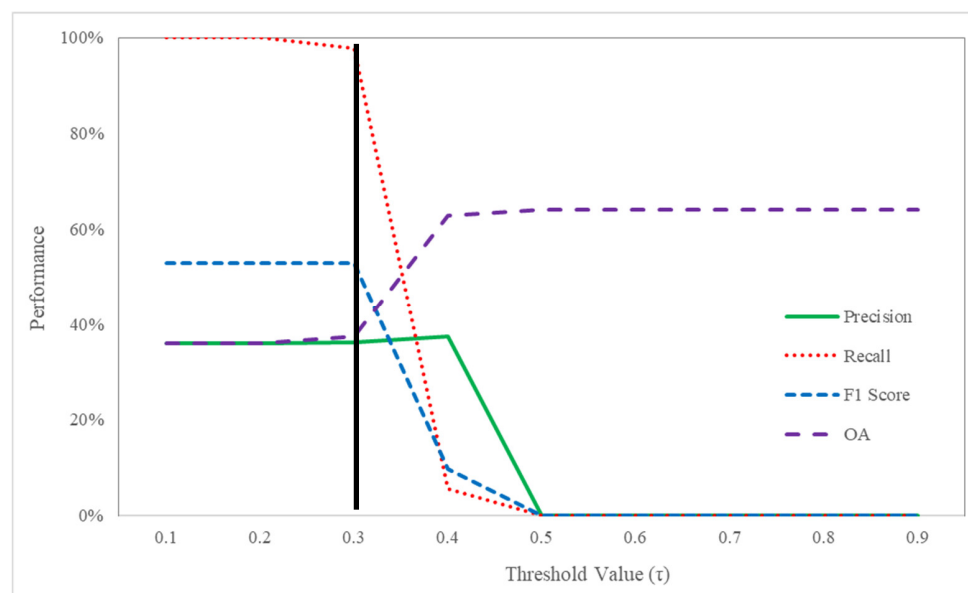
### 5.8. Artificial Neural Network (ANN)

A multilayer feed-forward artificial neural network (ANN) is the fifth classifier tested in this research. ANNs are highly interconnected networks that have robust computational and pattern recognition capabilities [26]. This machine learning model has also been widely used due to its ability to model non-linear relationships in a non-complex way. ANNs were originally developed to emulate the human brain, which has led to a growing usage of ANNs in the engineering field for numerical and statistical methods, in lieu of traditional classification and regression techniques. In general, one hidden layer is typically sufficient for solving most of the non-linear problems without network overfitting [26].

For this model, we used the Matlab Classification Learner application [23]. Only one hidden layer was used. In addition, the output layer was set to the softmax activation function, and the input and output data were standardized. This is typically recommended for ANNs so that the link weights and biases do not become extremely large or small [26]. The hyperparameters that were tuned were the number of hidden layer neurons and the activation function for the hidden layer neurons, both of which were performed using the random grid search algorithm and fivefold cross validation. The optimal number of hidden layer neurons was 15. The activation functions that were tuned were the hyperbolic tangent, sigmoid, rectified linear unit (ReLU), and no-activation function. The optimal activation function, which yielded the best results on the validation data, was the hyperbolic tangent function for the hidden layer neurons.

The trained ANN model was then applied to the unseen test dataset. This was carried out so that the optimal threshold value,  $\tau$ , may be optimized for the test data and so that a comparative evaluation could be made with the other machine learning algorithms on the basis of the evaluation metrics (Equations (1)–(4)). The performance curve results are presented in Figure 6.

Like the trained RF model, a majority of the ANN model's predictions for the unseen test data fell within the range of [0.3, 0.4], which is illustrated by drastic increases and decreases in the performance curves between these two threshold values. Since the recall is nearly perfect when using a threshold,  $\tau = 0.3$ , that threshold will be used as part of the comparative evaluation.

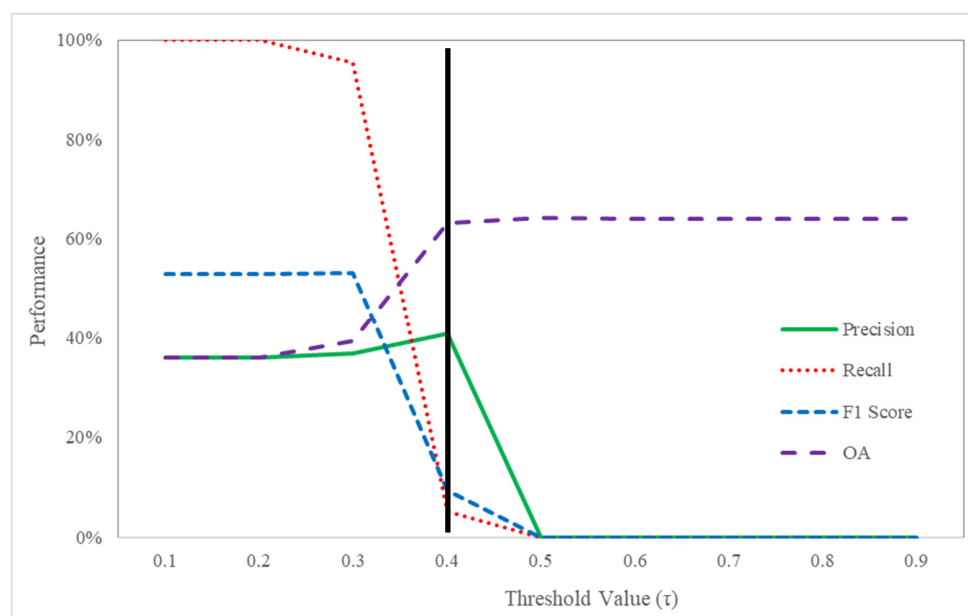


**Figure 6.** ANN performance curves from the test dataset.

### 5.9. Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is the sixth classifier tested in this research. It is a tree boosting method under the gradient boosting framework. XGBoost has been used for previous studies for classification tasks due to its good performance. Python programming was used for XGBoost model training and testing using the package sklearn [27,28]. The grid search was used to improve the XGBoost model performance by tuning the hyperparameters of number of estimators (50, 100, 200), maximum depth (3, 5, 7), and learning rate (0.001, 0.01, 0.1). The results show that XGBoost with a 0.01 learning rate, maximum depth of 3, and 200 estimators achieved the best performance.

In addition, the model's performance was evaluated on the test dataset. The threshold value,  $\tau$ , was optimized to achieve the best performance. Figure 7 shows the XGBoost model performance on the test dataset in terms of precision, recall, F1 score, and OA, as described in Equations (1)–(4). The results indicate that  $\tau = 0.4$  has the best performance in terms of its high precision and overall accuracy.



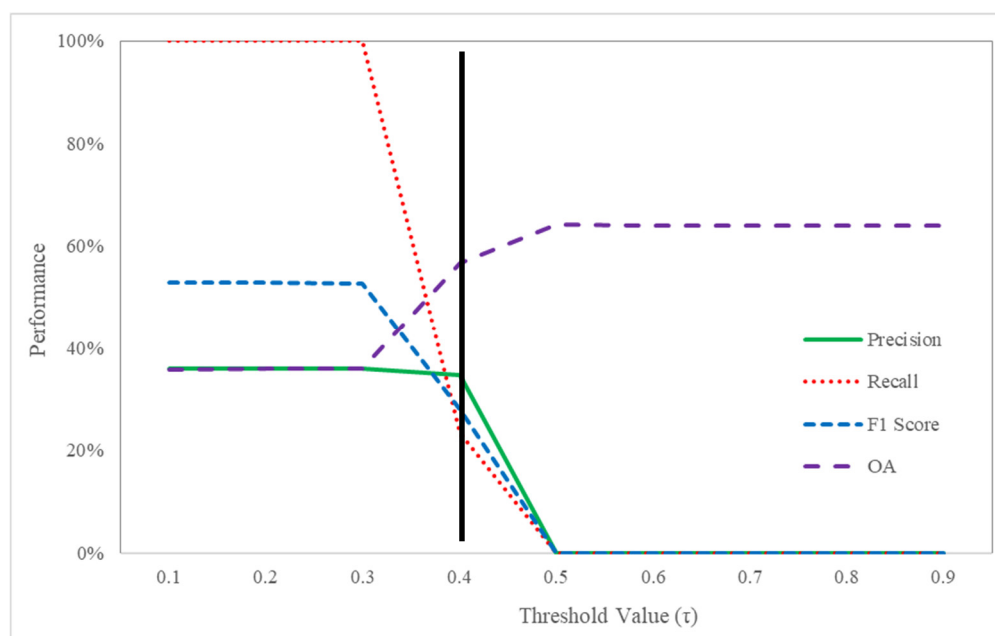
**Figure 7.** XGBoost performance curves from the test dataset.

### 5.10. Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is the seventh classifier tested in this research. The LSTM model is a type of recurrent neural network (RNN) model that can avoid the vanishing and exploding gradients, which are faced by the RNN model. The gating mechanism of the LSTM model enables it to solve the long-term decency problem when compared to the ANN model [27].

In this study, Python programming and the sklearn package were used to train and test the LSTM model [28]. The hyperparameters of the LSTM model were tuned by random search to improve its performance. The model's hyperparameters include nodes per layer, activation function, and learning rate. The range of the nodes per layer was from 32 to 128, with 16 as the increasing step; the choices for the activation functions were the rectified linear unit (ReLU) and hyperbolic tangent; and the range for the learning rate was 0.01, 0.001, and 0.0001. The random search results suggest that 32 nodes, with the hyperbolic tangent as the activation function, and a 0.0001 learning rate achieved the best performance.

The LSTM model's performance was also evaluated on the test dataset. Figure 8 shows the four evaluation metric changes with the threshold values. The results indicate that  $\tau = 0.4$  has the best performance in terms of both high OA and precision.



**Figure 8.** LSTM performance curves from the test dataset.

### 5.11. Comparative Evaluation

After optimizing the threshold values for each model, their performance could then be comparatively evaluated using the four performance metrics (Equations (1)–(4)). Table 5 summarizes the overall performance of the seven machine learning techniques evaluated.

Based on the results presented in Table 5, the RF and ANN models both yield the best and the same performance measures when using the optimal threshold value on unseen data. Although the BLR, KNN, NB, XGBoost, and LSTM models had similar precision scores, their recall scores were much lower than those of the RF and ANN models. This can also be seen by the lower F1 scores, as the F1 score is the harmonic mean between precision and recall. Because of this, the BLR, KNN, NB, XGBoost, and LSTM models' performances are too poor to be acceptable, and the RF or ANN models should be selected. In order to compare the results further, the confusion matrices of both the RF and ANN models are presented in Table 6 and Table 7, respectively.

**Table 5.** Comparative evaluation summary.

| Machine Learning Technique | Optimal Threshold Value ( $\tau$ ) | Precision (1) | Recall (2) | F1 Score (3) | OA (4) |
|----------------------------|------------------------------------|---------------|------------|--------------|--------|
| BLR                        | 0.3                                | 0.37          | 0.54       | 0.44         | 0.40   |
| KNN                        | 0.3                                | 0.37          | 0.50       | 0.42         | 0.47   |
| NB                         | 0.3                                | 0.39          | 0.46       | 0.42         | 0.55   |
| RF                         | 0.3                                | 0.36          | 0.98       | 0.53         | 0.38   |
| ANN                        | 0.3                                | 0.36          | 0.98       | 0.53         | 0.38   |
| XGBoost                    | 0.4                                | 0.41          | 0.05       | 0.09         | 0.63   |
| LSTM                       | 0.4                                | 0.35          | 0.24       | 0.28         | 0.57   |

**Table 6.** RF confusion matrix on the test dataset.

|                |           | $MD = 1$      | $MD = 0$      | Total | Correct Model Decision ( $MD \mid CS$ ) |
|----------------|-----------|---------------|---------------|-------|-----------------------------------------|
| Crash Severity | $CS = 1$  | 735           | 18            | 753   | 0.976 (97.6%)                           |
|                | $CS = 0$  | 1283          | 64            | 1347  | 0.048 (4.8%)                            |
| Total          |           | 2018          | 82            | 2100  | 0.380 (38.0%)                           |
| Precision      | $CS = MD$ | 0.364 (36.4%) | 0.780 (78.0%) |       |                                         |

**Table 7.** ANN confusion matrix on the test dataset.

|                |           | $MD = 1$      | $MD = 0$      | Total | Correct Model Decision ( $MD \mid CS$ ) |
|----------------|-----------|---------------|---------------|-------|-----------------------------------------|
| Crash Severity | $CS = 1$  | 736           | 17            | 753   | 0.977 (97.7%)                           |
|                | $CS = 0$  | 1296          | 51            | 1347  | 0.038 (3.8%)                            |
| Total          |           | 2032          | 68            | 2100  | 0.380 (38.0%)                           |
| Precision      | $CS = MD$ | 0.362 (36.2%) | 0.750 (75.0%) |       |                                         |

According to the results shown in Tables 6 and 7, the RF and ANN models are quite similar. Both models make almost identical predictions with regards to the severe crashes (i.e., recall values) at almost 98%; however, both models also conversely perform poorly when predicting non-severe crashes. This essentially means that both models make many [ $MD = 1$ ] decisions. This can be seen by the very high values in the [ $MD = 1$ ] column. The model that makes less [ $MD = 1$ ] incorrect decisions is the RF model. Therefore, the  $RF_{\tau=0.3}$  model, with 63 learners and four splits per learner, should be used to predict the severity of nighttime crashes based on roadway geometry, traffic volumes, weather characteristics, and lighting condition. This answers the first research question regarding the accuracy of a model to predict the severity of nighttime crashes.

## 6. Illumination Data Collection Tool

The third research question asked if a tool (e.g., a smartphone application) can be developed to accurately measure roadway illumination. Based on the BLR results from RQ1 and RQ2, the presence of illumination is close to being statistically significant, although it is not statistically significant at  $\alpha = 0.05$ . In addition, the coefficient for illumination was  $-0.089488$ , which makes sense intuitively, in that if illumination is added, the severity of a crash decreases.

In order for department of transportation (DOT) officials to use the recommended RF model, all of the input parameters must be readily available so that severe crash hot

spots may be predicted in the future. All of the input parameters are easily accessible to DOT officials through their roadway databases and a weather monitor. The other input parameter that is not readily available is the light condition (i.e., illumination on a roadway). Therefore, to answer RQ3, the research team is currently developing a smartphone application that makes use of a smartphone's global positioning system (GPS) coordinates, as well as the onboard camera, to record lux readings. These lux readings may then be imported into a geographic information system (GIS) to determine the locations of good and poor illumination.

Since the application is still under development, the research team conducted a pilot study at The University of Texas at Tyler (UT Tyler) main campus using a lux meter. The lux meter MT-912 [29] was used to collect lux readings at night at 85 different locations on campus. Table 8 presents the descriptive statistics for the collected data.

**Table 8.** Descriptive statistics of the pilot study.

| Location       | Within UT Tyler Campus            | Adjacent to UT Tyler Campus       |
|----------------|-----------------------------------|-----------------------------------|
| Unit           | Lux $\left(\frac{lm}{m^2}\right)$ | Lux $\left(\frac{lm}{m^2}\right)$ |
| Sample size    | 76                                | 9                                 |
| Min            | 0.0                               | 0.0                               |
| Max            | 73.7                              | 10.7                              |
| Mean           | 20.7                              | 2.4                               |
| Std. Deviation | 15.7                              | 3.9                               |

The collected data were then imported into ArcGIS [30] to develop a heatmap of the illumination intensity at the UT Tyler main campus. Figure 9 presents the illumination heatmap.



**Figure 9.** Pilot study illumination heatmap of UT Tyler main campus.

Based on the results presented in Figure 9, the UT Tyler decision makers should consider adding illumination infrastructure (i.e., light poles) at the locations shown in dark green, as these are the places that lack lighting the most.

Once the smartphone application has been developed, it would follow the same methodology of using a smartphone's onboard camera to record lux readings in addition to the geolocation (i.e., latitude and longitude) so that a similar heatmap may be made of the city, TxDOT district, and/or at state level. The crowdsourced illumination data would theoretically be collected at a cost that approaches zero for the DOTs.

## 7. Conclusions

The overall objective of this research was to demonstrate the importance of roadway illumination with respect to crash severity. As such, the research team used crash data from the TxDOT CRIS database [15], specifically for nighttime crashes in Smith County, Texas (i.e., county of Tyler, Texas), from the period of 2013–2023. The data were split into a training dataset to develop machine learning models, and then the models were comparatively evaluated based on their performance using the test dataset. Based on the comparative evaluation results, the  $RF_{\tau=0.3}$  model, with 63 learners and four splits per learner, should be used to predict the severity of nighttime crashes due to its high recall value of 97.6% when predicting severe crashes. This recommended RF model may be used by DOTs to predict the severity of nighttime crashes using only readily available roadway geometry, traffic volume, weather characteristics, and illumination data.

RQ2 asked if the presence of illumination is significant with respect to nighttime crash severity. Based on the BLR results, the light condition parameter had a *P* value of approximately 0.123. While this is not high enough to be considered statistically significant at  $\alpha = 0.05$ , it could still be interpreted as providing some evidence against the null hypothesis. The coefficient for the illumination parameter, on the other hand, was  $-0.089488$ , which makes sense intuitively given that, if illumination is added, the severity of crashes will decrease.

Finally, RQ3 asked if a tool can be developed to assist DOT decision makers in collecting illumination data so as to prioritize funding for poorly illuminated crash hot spots. The research team is currently developing a smartphone application that collects GPS location and illumination data through the sensors on a smartphone. In the interim, a pilot study was conducted using a handheld lux meter.

As with any research, there are limitations to this study. The limitations are as follows:

- For this research, traditional classifiers such as ensemble methods and ANNs were used. In addition, two more sophisticated models were used: XGBoost and LSTM. Future research should include even more sophisticated methods, such as logit models that incorporate random parameters with heterogeneity in means and variances, both of which are commonly used.
- The data were only collected in Smith County, Texas (i.e., the county where Tyler, Texas, is located). It is not clear how the RF model will perform on roadways outside of Smith County, Texas.
- The time period for these crashes was from 2013 to 2023, which includes temporal instability, particularly due to the COVID-19 pandemic. This has not been addressed as part of this research.

For future research, we plan to achieve the following: (i) explore additional machine learning techniques to determine if they can surpass the performance of the RF model developed in this study; (ii) gather actual illumination data using a lux meter and/or smartphone devices to include as one of the input parameters; and (iii) create a model to predict the number of crashes rather than their severity. If the RF model developed

in this study outperforms additional machine learning techniques, we plan on testing its robustness with missing data as part of a fault tolerance analysis similar to past work that has been performed on fault tolerance analyses in transportation engineering [16].

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